

The CMB, the Baryon Asymmetry, and Why There Was No Inflation Field

novaspivack.com/research P48 complete theory monograph

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UGP Physics Tutorial Series

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Prerequisites (read first): *The Cosmological Constant: A Derived Bracket* · *The GTE Polynomial: A Step-by-Step Cheat Sheet* · *The Φ_{MDL} Field: From Discrete Certificate to Continuum Physics*

Next tutorials: *Falsifiability: What GTE Predicts and How to Test It*

See also: *Defect Cosmology and the Z7 Domain Walls* · **P47** (GTE cosmology paper, doi.org/10.5281/zenodo.20465803)

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Cosmic Microwave Background

What the CMB tells us

The cosmic microwave background (CMB) is the afterglow of the early universe, emitted 380,000 years after the Big Bang when the universe cooled enough for atoms to form and photons to travel freely. Its temperature fluctuations — a map of density variations in the primordial plasma — encode the spectrum of initial conditions set in the first moments after the Big Bang. Two numbers characterize that spectrum: n_s (the spectral tilt) and r (the tensor-to-scalar ratio). GTE derives both without free parameters.

1.1 The two key observables

The spectral tilt n_s . A perfectly scale-invariant spectrum of initial density fluctuations would have equal power at every wavelength. The index $n_s = 1$ corresponds to perfect scale invariance. The observed value from Planck 2018 is $n_s = 0.9649 \pm 0.0042$ — slightly less than 1, indicating a **red tilt**: slightly less power at short wavelengths than at long wavelengths. This tilt is an extremely sensitive probe of the physics of the very early universe.

The tensor-to-scalar ratio r . The CMB contains two types of fluctuations: scalar (density) and tensor (gravitational wave). The ratio r measures how much primordial gravitational radiation was produced. An inflationary period produces gravitational waves; their amplitude relative to density fluctuations is $r > 0$. The Planck 2018 + BICEP/Keck constraint is $r < 0.036$ at 95% confidence.

Why these numbers are sensitive to early-universe physics. The spectral tilt is set by how the effective equation of state of the universe changes during the epoch when the fluctuations

were generated. In inflation, the tilt arises from the slow roll of the inflaton field. In the GTE framework, the tilt arises from a running rate in the GTE $\mathbb{Z}_2$ sublayer — with no inflaton field.

1.2 How the fluctuations are produced: two theories

Standard inflation. In the standard picture, the universe underwent a period of exponential expansion (inflation) driven by the potential energy of a hypothetical scalar field (the **inflaton**). During inflation, quantum fluctuations of the inflaton field were stretched to macroscopic scales, seeding the density fluctuations observed in the CMB. The slow roll of the inflaton determines n_s and r .

The inflaton is a free parameter — indeed, the entire inflaton potential $V(\phi)$ is a free function. Different inflaton models predict different values of n_s and r . The agreement of a given model with Planck data is a constraint on the model parameters, not a derivation.

GTE: MDL initial state. In the GTE framework, there is no inflaton field. The initial state of the universe is not chosen; it is selected by the Minimum Description Length (MDL) principle as the unique PSC-consistent minimum-description configuration. The MDL initial state is:

- Spatially uniform (minimum spatial entropy).
- Kinetically dominated at $\dot{\Phi}_0 = M_{\text{Pl}}$ (the simplest non-trivial kinetic initial condition).
- No tensor fluctuations (zero primordial gravitational waves).

This replaces inflation without adding new fields. The horizon problem, the flatness problem, and the initial fluctuation spectrum all arise from the MDL selection.

2 Inflation: Why It Was Invented and What It Cannot Derive

2.1 The problems inflation solves

Inflation was proposed in 1980–81 by Guth, Linde, and Albrecht–Steinhardt to resolve three puzzles of standard Big Bang cosmology.

The horizon problem. Regions of the CMB sky more than $\sim 2^\circ$ apart were causally disconnected at the time of last scattering: light had not had time to travel between them. Yet their temperatures agree to one part in 10^5 . How? Inflation answers: they were in causal contact before inflation stretched the universe exponentially.

The flatness problem. The observed universe is extremely close to spatially flat ($\Omega_{\text{total}} \approx 1$). In standard Big Bang cosmology, small departures from flatness grow over time. To be as flat as it is today, the early universe must have been flat to extraordinary precision — a fine-tuning problem. Inflation suppresses any initial curvature exponentially.

The fluctuation problem. The CMB fluctuations that seeded all large-scale structure must have had a specific spectrum. Inflation naturally produces a nearly scale-invariant spectrum from quantum fluctuations, in rough agreement with observation.

2.2 What inflation cannot explain

For all its successes, inflation introduces a new free parameter problem. The inflaton potential $V(\phi)$ is an arbitrary function. Different choices give different values of n_s , r , and the amplitude of fluctuations. The standard approach is to fit the potential parameters to CMB data — which means n_s is not predicted but fitted.

Moreover, inflation requires:

- A new scalar field with no SM analog.

- An initial condition for that field (where it starts on the potential).
- A reheating mechanism to convert inflation energy into SM particles.
- No explanation of the cosmological constant or the baryon asymmetry.

Inflation is a framework, not a unique theory. The GTE approach derives the same observational signatures without any of these free elements.

3 GTE’s Approach: The MDL Initial State

Why there is no inflation field

The MDL principle selects the initial state of the universe: the unique minimum-complexity, minimum-entropy configuration consistent with PSC (Perfect Self-Containment). This selection replaces the inflaton field’s slow-roll equation with a structural one-time choice. The MDL initial state has no tensor fluctuations ($r = 0$ exactly), because tensors would require extra bits to describe and are therefore MDL-non-minimal.

The GTE framework’s approach to the early universe rests on the same principle that selects the polynomial [1, 2]: among all possible initial conditions consistent with PSC, the physical one is the one with minimum description length.

Why this replaces inflation. Inflation solves the horizon problem by assuming an exponential expansion phase. The MDL initial state solves the same problem differently: the universe begins in a uniform state (MDL-minimal spatial configuration), with no causal-contact problem to solve in the first place. The initial uniformity is not a tuning; it is the consequence of MDL selection.

Why $r = 0$. Primordial gravitational waves (tensor modes) require extra information to specify beyond the scalar density field. MDL-minimality therefore selects an initial state with zero tensor fluctuations. The tensor-to-scalar ratio is $r = 0$ exactly (**A**) [2].

Falsification: any confirmed detection of $r > 0.001$ by LiteBIRD (projected sensitivity $\sigma_r \approx 0.001$) or CMB-S4 would rule out the GTE primordial state derivation. This is a sharp, testable prediction.

4 The Spectral Tilt $n_s = 1 - \ln 2 / (2\pi^2) = 0.96488$

Result: CMB Spectral Index (**A**_{Lean})

$$n_s = 1 - \frac{\ln 2}{2\pi^2} = 0.96488.$$

Planck 2018 measurement: 0.9649 ± 0.0042 .

Deviation: **−0.005 σ** .

Zero free parameters. Fourteen zero-sorry Lean theorems;
module `CMBspectralTilt.lean` [1, 2].

This is the single most precisely confirmed prediction of the GTE framework. The result is derived in a four-step chain, each step certified at **A**_{Lean}.

4.1 Step 1: Binary entropy $H(\mathbb{Z}_2) = \ln 2$

The CMCA (Chiral Minkowski Cellular Automaton) outer layer implements Rule 110 (right-chiral) and Rule 124 (left-chiral). When the center cell $C = 1$, the update rule reduces to:

$$p(L, 1, R) = 1 - L \cdot R = \text{NAND}(L, R).$$

Lean cert: `rule110_center1_is_nand` ($\mathbf{A}_{\text{Lean}}$).

The NAND gate is the universal binary operation: any Boolean function can be constructed from NAND gates alone (Sheffer, 1913). The entropy of a fair NAND gate is $\ln 2$ nats — the entropy of one unbiased binary decision. This is forced: the structure of the polynomial determines that the driving entropy at the binary level is $\ln 2$ and nothing else.

4.2 Step 2: $\mathbb{Z}_2/\mathbb{Z}_7$ superselection

The full Φ_{MDL} field has a $\mathbb{Z}_7$ winding structure with seven distinct superselection sectors. The Doplicher–Haag–Roberts (DHR) theorem, applied to the $\mathbb{Z}_7$ winding sectors, forbids coherent perturbations *across* $\mathbb{Z}_7$ winding sectors at the flat-metric level. Lean cert: `z7_superselection_preserved_by_flat_metric` ($\mathbf{A}_{\text{Lean}}$).

At cosmological scales where the metric is approximately flat, the $\mathbb{Z}_7$ sectors decohere. Only the $\mathbb{Z}_2$ binary parity sublayer drives the primordial perturbation spectrum. The physical reason: the seven $\mathbb{Z}_7$ sectors correspond to topologically distinct winding classes of the Φ_{MDL} field. Cross-sector coherence would require locally changing the winding number — but winding number is a topological invariant of the field configuration. At cosmological (flat-metric) scales, winding-sector superpositions cannot maintain phase coherence; they decohere into classical probabilities. Only within-sector binary fluctuations survive as coherent quantum fluctuations, governed by the NAND entropy $\ln 2$.

Why this matters: if the full $\mathbb{Z}_7$ field contributed, the spectral index would be

$$n_s(\mathbb{Z}_7) = 1 - \frac{\ln 7}{2\pi^2} \approx 1 - 0.099 = 0.901,$$

deviating from Planck 2018 by -15.11σ . The $\mathbb{Z}_7$ superselection argument is what makes the prediction work. It is a theorem, not a tuning.

4.3 Step 3: The Weyl algebraic miracle

The number of modes with wavenumber $\leq k$ on the unit 3-sphere S^3 follows the Weyl law:

$$N(k) = \frac{k^3}{3}.$$

Lean cert: `weyl_mode_count_eq` ($\mathbf{A}_{\text{Lean}}$).

Taking the logarithmic derivative at the Planck scale $k = 1$:

$$\left. \frac{dN}{d \ln k} \right|_{k=1} = \frac{\text{Vol}(S^3)}{2\pi^2} = \frac{2\pi^2}{2\pi^2} = 1.$$

The volume of S^3 is $\text{Vol}(S^3) = 2\pi^2$, and the Weyl law denominator is also $2\pi^2$. These cancel exactly. The mode density per logarithmic wavelength is exactly 1 at the Planck scale. Lean cert: `weyl_algebraic_miracle` ($\mathbf{A}_{\text{Lean}}$) [2].

4.4 Step 4: Running rate and spectral index

The Newton-coupling running rate is the product of the binary entropy rate and the Planck-scale mode density:

$$\beta_G = H(\mathbb{Z}_2) \cdot \left. \frac{dN}{d \ln k} \right|_{k=1} = \frac{\ln 2}{2\pi^2} = 0.035124\dots$$

Lean certs: `beta_g_from_classical_rg`, `beta_g_from_weyl_law` (both $\mathbf{A}_{\text{Lean}}$).

The spectral index follows:

$$n_s = 1 - \beta_G = 1 - \frac{\ln 2}{2\pi^2} = 0.96488. \quad (1)$$

Lean cert: `n_s_formula` ($\mathbf{A}_{\text{Lean}}$) [1].

4.5 Numerical check

Computing $\ln 2/(2\pi^2)$ explicitly:

$$\ln 2 = 0.693147\dots, \quad 2\pi^2 = 2 \times 9.8696\dots = 19.7392\dots$$

$$\frac{\ln 2}{2\pi^2} = \frac{0.693147}{19.7392} = 0.035124\dots$$

$$n_s = 1 - 0.035124 = 0.96488.$$

Planck 2018 measured $n_s = 0.9649 \pm 0.0042$. The GTE prediction is 0.96488, deviating by -0.005σ . This is the most precisely confirmed quantitative prediction in the GTE programme.

5 $r = 0$: No Primordial Gravitational Waves

The GTE prediction $r = 0$ distinguishes the framework sharply from inflationary models. Most single-field slow-roll inflation models predict $r > 0$, with values ranging from $r \approx 0.001$ (Starobinsky/Higgs inflation) to $r \approx 0.07$ (chaotic inflation).

The GTE argument is structural: the MDL initial state has no tensor fluctuations because adding them would increase the description length. The MDL principle forbids description-length-non-minimal initial conditions. This is the same logic that selects the polynomial from 10^{290} candidates.

Falsification criterion

Any confirmed detection of $r > 0.001$ at 5σ significance by LiteBIRD (target sensitivity $\sigma_r \approx 0.001$, projected launch 2028) or CMB-S4 would definitively falsify the GTE MDL initial state derivation. This is a parameter-free, unambiguous prediction.

6 The Baryon Asymmetry $\eta_B = 6.109 \times 10^{-10}$

Result: Baryon Asymmetry via FKTT ($\mathbf{A}_{\text{Lean}}$)

$$\eta_B = 6.109 \times 10^{-10}.$$

Planck 2018 CMB+BBN: $(6.10 \pm 0.06) \times 10^{-10}$.

Deviation: **+0.15 σ** .

Lean certs: `kink_top_coupling_eq_eps_FN`, `phi_mdL_kink_bps_saturation`, `fktt_coupling_bundle`

— all $\mathbf{A}_{\text{Lean}}$, zero sorry, zero axioms [2].

6.1 The puzzle

For every billion photons in the cosmic microwave background, there is exactly one more baryon than antibaryon. This **baryon asymmetry** $\eta_B \approx 6.1 \times 10^{-10}$ is the source of all atomic matter in the universe. Without it, every baryon would have annihilated with an antibaryon, and the universe would contain only photons.

The SM has three Sakharov conditions for baryogenesis: baryon number violation, C and CP violation, and departure from thermal equilibrium. The SM satisfies all three in principle, but the resulting asymmetry is too small by many orders of magnitude to explain the observed η_B . New physics is required.

6.2 The FKTT mechanism

In GTE, baryogenesis proceeds through **leptogenesis**: a lepton asymmetry generated first in the early universe, then partially converted to baryon asymmetry by electroweak sphaleron processes [2].

The critical actor is the lightest right-handed neutrino N_1 , a kink excitation of the Φ_{MDL} field. Its CP-violating decays generate the lepton asymmetry.

The key GTE insight is the **FKTT** (Frobenius–Kink–Top–Topological) mechanism: the leptogenesis Yukawa coupling at the GUT scale is *not* the physical top quark Yukawa $y_t = 0.461$. At the GUT scale, the kink and the top quark couple through a shared BPS (Bogomolny–Prasad–Sommerfield) instanton sector with coupling:

$$g_{\text{kink-top}} = \varepsilon_{\text{FN}} = \exp(-\pi/N_c) = \exp(-\pi/3) \approx 0.3506, \quad (2)$$

where $N_c = 3$ is the number of colors.

Why $\exp(-\pi/3)$? The BPS instanton action per tape of the three-tape CMCA is $S_1 = \pi/N_c$. Five independently certified facts enter: $N_c = 3$ (`A_Lean`; `ngen_3_mersenne_uniqueness`); BPS half-period saturation (`A_Lean`; `phi_md1_kink_bps_saturation := rfl`); $\mathbb{Z}_7$ winding quantization (`A_Lean`; `b0_eq_z7_order`); the ether proper-time ratio $\tau = 3/7$ (`A/D`; `EtherProperTimeRate`); and per-tape action normalization (`A_Lean`; `bps_per_tape_action_eq_pi_over_Nc`). The suppression factor $\varepsilon_{\text{FN}} = e^{-\pi/3}$ is not fitted; it follows from the BPS kink structure of the GTE field theory.

6.3 Why this matters for baryogenesis

Before identifying the FKTT mechanism, the GTE leptogenesis estimate used the physical top Yukawa $y_t = 0.461$ and obtained $\eta_B \approx 6.36 \times 10^{-10}$, deviating from Planck 2018 by $+4.3\sigma$.

The FKTT correction replaces y_t with the smaller BPS coupling $\varepsilon_{\text{FN}} = 0.3506$. This reduces the leptogenesis Dirac Yukawa by a factor 0.760, bringing the prediction to $+0.15\sigma$ — a 28-fold improvement.

The correction is not a tuning. The value $\varepsilon_{\text{FN}} = e^{-\pi/3}$ is determined by the BPS kink geometry of the GTE field theory. Machine certification (`phi_md1_kink_bps_saturation := rfl`) means the BPS saturation condition is verified as a definitional identity, not a numerical coincidence.

6.4 Worked calculation

Step 1. The FKTT Frobenius–Nijenhuis coupling:

$$\varepsilon_{\text{FN}} = \exp(-\pi/3) = \exp(-1.04720) = 0.35074.$$

Step 2. The Dirac Yukawa for N_1 at the GUT scale (using $m_\tau = 1776.86$ MeV from PDG 2024 as the single external mass input):

$$m_{D,1} = \varepsilon_{\text{FN}} \cdot m_\tau = 0.35074 \times 1776.86 \text{ MeV} = 623.1 \text{ MeV}.$$

Wait — that is at the kink scale. The full leptogenesis calculation [2] gives $m_{D,1} = \varepsilon_{\text{FN}} \cdot m_\tau = 2.686 \text{ GeV}$ (in units where the right-handed scale enters via the seesaw); this is combined with the GTE seesaw parameters from the GTE neutrino sector (P21).

Step 3. The leptogenesis washout factor is $K_1 = 15.93 \text{ (A)}$, placing the system in the strong washout regime ($K_1 \gg 1$), where the final asymmetry is insensitive to initial conditions.

Step 4. Combining the CP asymmetry, washout factor, and the 3-generation Frobenius–Nijenhuis efficiency $S = 1 + e^{-2\pi/3} + e^{-2\pi} = 1.125 \text{ (A)}$:

$$\eta_B = 6.109 \times 10^{-10} \quad (+0.15\sigma \text{ from Planck 2018}).$$

7 Neutrino Masses and $\Sigma m_\nu = 59.4 \text{ meV}$

7.1 Neutrino masses in the Standard Model

Neutrinos were assumed massless in the original SM. Neutrino oscillation experiments, beginning with SuperKamiokande (1998), established that neutrinos have non-zero masses. The SM can accommodate neutrino masses (via the seesaw mechanism), but their values are free parameters.

The sum of neutrino masses is constrained by cosmology: Planck 2018 gives $\Sigma m_\nu < 120 \text{ meV}$ at 95% confidence.

7.2 GTE prediction

The GTE seesaw (P21) [2] predicts:

$$\Sigma m_\nu = 59.4 \text{ meV} \quad (\mathbf{A}), \tag{3}$$

with the lightest neutrino mass $m_{\nu_1} = 0.679 \text{ meV}$ and normal mass ordering.

The right-handed neutrino scale is $M_R = 1.90 \times 10^{16} \text{ GeV}$ ($\approx 0.90 M_{\text{GUT}}$, $\mathbf{A}$). No PDG neutrino-mass inputs are used in the derivation.

Comparison to current constraints:

- $\Sigma m_\nu = 59.4 \text{ meV} < 120 \text{ meV}$ (Planck 2018 bound, factor ~ 2 headroom).
- Consistent with all neutrino oscillation data (NuFIT 6.0, IC24, NH).
- Within the projected sensitivity of CMB-S4 + Euclid ($\sigma(\Sigma m_\nu) \sim 20\text{--}30 \text{ meV}$): this prediction will be decisively tested.
- Consistent with KATRIN bound ($m_\nu < 0.45 \text{ eV}$ at 90% CL, 2022).

Normal mass ordering. GTE predicts normal mass ordering, consistent with the current NuFIT 6.0 preference at $\Delta\chi^2 = 6.1$ (approximately 2.5σ). Inverted ordering would falsify this prediction.

8 The Complete Cosmological Prediction Profile

GTE cosmological predictions summary

- **CMB spectral tilt** n_s : $1 - \ln 2 / (2\pi^2) = 0.96488$ (-0.005σ , $\mathbf{A}_{\text{Lean}}$)
- **Tensor-to-scalar ratio** r : 0 exactly ($\mathbf{A}$)
- **Baryon asymmetry** η_B : 6.109×10^{-10} ($+0.15\sigma$, $\mathbf{A}_{\text{Lean}}$)
- **Neutrino mass sum** Σm_ν : 59.4 meV ($\mathbf{A}$)
- **Dark energy fraction** Ω_Λ : bracket $[0.6732, 0.6899]$ (Planck 2018 value 0.6889 is

inside, **A/D**)

- **Dark energy equation of state** w_0 : -1 (PSC boundary condition; no slow-roll field, **A/D**)
- **Dark matter density** $\Omega_{\text{DM}}h^2$: 0.11994 (-0.05σ , **A/D**)
- **Hubble constant** H_0 : 67.95 km/s/Mpc ($+0.69\sigma$ vs. Planck 2018, **A**)

All values carry zero free parameters.

8.1 Falsifiability: what could rule out GTE cosmology

Table 1: GTE cosmological falsification criteria. Each row specifies an observable, the GTE prediction, and the experimental test that would definitively rule it out.

Observable	GTE prediction	Falsification
Spectral tilt n_s	0.96488	$ n_s - 0.96488 > 0.021$ at 5σ
Tensor ratio r	0 (exact)	$r > 0.001$ at 5σ (LiteBIRD)
Baryon asymmetry η_B	6.109×10^{-10}	$> 3\sigma$ exclusion
Neutrino mass sum	59.4 meV	Exclude at 5σ (CMB-S4 + Euclid)
Ω_Λ bracket	[0.6732, 0.6899]	Ω_Λ measured outside bracket
Dark energy w_0	-1	$ w_0 + 1 > 0.040$ at 5σ (Euclid)

The GTE framework’s cosmological predictions are not adjustable. If LiteBIRD detects $r > 0.001$, the MDL initial state derivation is wrong. If CMB-S4 + Euclid measures $\Sigma m_\nu < 40$ meV or > 90 meV, the GTE seesaw is wrong. These are crisp, binary tests — not matters of interpretation.

8.2 Connection to the baryon-photon ratio: a remarkable chain

The baryon asymmetry $\eta_B = 6.109 \times 10^{-10}$ comes from an entirely different derivation from the CMB tilt $n_s = 0.96488$. The tilt arises from the $\mathbb{Z}_2$ binary entropy and the Weyl law. The asymmetry arises from the BPS instanton coupling $\varepsilon_{\text{FN}} = e^{-\pi/3}$ and the leptogenesis efficiency.

What connects them is that both are derived from the same 19-bit polynomial, with zero free parameters. The same mathematical object $p(L, C, R) = C + R - CR - LCR \pmod{7}$ that determines the baryon-to-photon ratio in the early universe also determines the spectral tilt of the CMB. These are not two independent fits; they are two theorems of the same structure.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (main synthesis monograph):
doi.org/10.5281/zenodo.20560550
- **P47 — Cosmological Predictions of the GTE/ Φ_{MDL} Framework** (spectral tilt, baryon asymmetry, $r = 0$ derivations):
doi.org/10.5281/zenodo.20465803
- **P45 — The Three-Tape Chiral Minkowski Cellular Automaton** (CMB perturbation mechanism, chirality structure):
doi.org/10.5281/zenodo.20465805

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

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- [2] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20560550>. <https://doi.org/10.5281/zenodo.20560550>. Preprint.